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Assessment of readiness in the diagnosis and management of central disorders of hypersomnolence: A survey of sleep medicine program directors and fellowship graduates

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ABSTRACT

Objective/background: Central disorders of hypersomnolence (CDH), including narcolepsy, idiopathic hypersomnia, and recurrent hypersomnia, are rare conditions that pose challenges in sleep medicine fellowship training. Limited exposure to these disorders, variability in training experiences, and growing complexity of diagnostic tools and treatment options contribute to gaps in education. This study aimed to assess perceptions of training adequacy in CDH among sleep medicine program directors and recent fellowship graduates.
Patients/methods: A cross-sectional survey was conducted between April 2024 and July 2024 among US board-certified sleep physicians, including program directors contacted through a sleep medicine fellowship program listserv, and their recent fellowship graduates. Surveys assessed perceptions of training in diagnosing and managing CDH, including exposure to diagnostic tools, patient populations, and pharmacological treatments. Responses were summarized using descriptive statistics.
Results: Of 110 program directors contacted, 29 (26.4%) responded, and 48 (6.5%) of 740 eligible graduates participated. Program directors and graduates reported adequate exposure to narcolepsy and idiopathic hypersomnia; exposure to recurrent hypersomnia was lower. Both groups reported confidence in adequacy of training in most diagnostic methods (polysomnography, multiple sleep latency test, and Epworth Sleepiness Scale); confidence with actigraphy was lower. Gaps were identified in training for longitudinal care during medical management of CDH, and exposure to certain medications (eg, lithium) was limited.
Conclusions: Survey findings demonstrate adequacy of training in diagnosing CDH but suggest some gaps in training regarding rare conditions and advanced treatment and management, highlighting needs for targeted educational resources and standardized curricula.
Clinical trial registration: Not applicable.

1. Introduction

Despite advances in sleep medicine, significant challenges persist in sleep medicine fellowship training programs related to the diagnosis and management of central disorders of hypersomnolence (CDH; also known as hypersomnias), including narcolepsy, idiopathic hypersomnia, and recurrent hypersomnia (also known as Kleine-Levin syndrome) [1–4]. CDH are rare [4,5], resulting in limited training opportunities in this area of medicine. Across professional societies, sleep medicine educational resources may be difficult and cumbersome to access [6,7], may have financial barriers, or may be out of date.

The pathway to sleep medicine board certification requires a 1-year training program that involves completion of an Accreditation Council for Graduate Medical Education (ACGME)-accredited sleep medicine fellowship [1]. In the absence of a singular track to enter sleep medicine,

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Glossary:

AAN	American Academy of Neurology
ACGME	Accreditation Council for Graduate Medical Education
CDH	central disorders of hypersomnolence
ESS	Epworth Sleepiness Scale
IRB	Institutional Review Board
IQR	interquartile range
MSLT	Multiple Sleep Latency Test
NT1	narcolepsy type 1
NT2	narcolepsy type 2
PSG	polysomnography

prospective fellows come from a variety of medical specialties (including internal medicine, family medicine, anesthesia, otolaryngology, psychiatry, pediatrics, neurology, cardiology, and pulmonology). Consequently, fellows possess varying degrees of familiarity with the numerous aspects of sleep medicine. Fellowship training might also be geared toward board exam preparation, favoring examinable facts such as those regarding diagnosis or clinical definition over longitudinal treatment, which can discourage continuity of care.

National Resident Matching Program data reveal a growing number of sleep medicine fellowship programs and applicants to these programs, which may reflect an increasing interest in sleep medicine [8]. Given the growing complexity of hypersomnia education and clinical resources, there is heightened awareness that fellows should be introduced to this material early in their training and supported through ongoing clinical mentorship. Sleep medicine fellows need to be well versed in a wide and expanding array of diagnostic tools and diagnostic protocols, including the Epworth Sleepiness Scale (ESS), actigraphy, and sleep studies (nocturnal polysomnography [PSG], Multiple Sleep Latency Test [MSLT]) [9]. In addition, the number of recently approved medications and those in development for CDH continues to grow [3]. Historically, medications used for CDH have included traditional stimulants, high-sodium oxybate (beginning in the 1960s to treat daytime and nighttime symptoms, and approved for cataplexy in narcolepsy in 2002, and subsequently excessive daytime sleepiness [EDS] in 2005 [10,11]), modafinil (approved for EDS associated with narcolepsy in 1998 [12]), and armodafinil (approved for EDS associated with narcolepsy in 2007 [13]) [14]. More recent medication approvals in sleep medicine include wake-promoting agents (eg, pitolisant [15–17], approved for EDS associated with narcolepsy in 2019 and for cataplexy in 2020, and solriamfetol [18], approved for EDS associated with narcolepsy in 2019), low-sodium oxybate (approved for EDS or cataplexy in patients 7 and older with narcolepsy in 2020 and idiopathic hypersomnia in adults in 2021 [19]), and fixed-dose high-sodium oxybate (approved for EDS and cataplexy in adults with narcolepsy in 2023 and for patients ≥7 years of age in 2024 [20]) [14]. Additionally, American Academy of Sleep Medicine guidelines for the treatment of CDH include conditional recommendations for off-label use of interventions such as clarithromycin for idiopathic hypersomnia and lithium for recurrent hypersomnia [3]. The evolving treatment landscape over the past several years has introduced additional complexity in treatment selection for CDH.

In the setting of a 1-year training program, gaining comprehensive knowledge in the diagnosis and management of patients with CDH can be challenging, and graduating physicians may not feel fully prepared for independent management of these patients. The ACGME [21] sets minimum standards and requirements for sleep medicine training. ACGME guidance is deliberately designed to not be overly prescriptive, leading to substantial variability in training experiences, which may differ both between and within programs (from fellow to fellow and from year to year) based on patient populations, department and/or faculty specialties, institutional culture, and prescribing practices. Thus, training programs may comply with ACGME guidelines but not fully address the educational needs of sleep medicine fellows and program directors, particularly concerning rare disease conditions such as CDH.

There is an opportunity to complement education on CDH provided through standardized sleep medicine education. Jazz Sleep Academy™ is a structured initiative to advance collaboration, education, and clinical relevance among the sleep community to help optimize patient care. As part of this initiative, a descriptive, observational, cross-sectional survey was first developed to assess gaps in education on CDH among sleep medicine training program directors and recent sleep medicine fellowship graduates. Using data from this survey, we assessed preparedness for longitudinal management of these rare disorders by graduates of 1-year training programs.

2. Materials and methods

2.1. Development of Jazz Sleep Academy™

In 2021, the concept of Jazz Sleep Academy™ was raised by the sponsor's Medical Affairs department [22]. A series of advisory boards comprising clinicians, sleep specialists, and other experts was convened to obtain input and feedback on development of a survey to assess readiness of sleep medicine fellowship graduates in the diagnosis and management of CDH.

2.2. Survey design and objectives

A cross-sectional email survey of eligible US physicians was conducted to assess readiness and gaps in education in the diagnosis and management of CDH (narcolepsy, idiopathic hypersomnia, and recurrent hypersomnia) upon graduation from sleep medicine fellowship training programs from the perspective of the program directors and post-fellowship graduates. Eligible program director participants were US board-certified sleep physicians who were active sleep medicine fellowship program directors and associate program directors; eligible post-fellowship graduates were physicians who had completed a sleep medicine fellowship within the last 5 years. Eligibility was by self-report, with an anticipated sample size of 50 to 100 respondents in each cohort. Former sleep medicine program directors and physicians who had completed their sleep medicine fellowship program more than 5 years ago were excluded, as were trainees who were actively in a sleep medicine fellowship program.

Approximately 110 program directors (including associate program directors) were contacted through a sleep medicine fellowship program listserv for survey participation and asked to forward the survey to their graduates who finished the program within the past 5 years. Trainees and fellows who recently graduated were also identified for survey participation through sponsor contact information lists based on participation in educational efforts and professional society channels focusing on early career channels (eg, American Academy of Sleep Medicine Engage platform, American Academy of Neurology [AAN] Synapse platform).

Two web-based surveys were created by 2 members of the Jazz Sleep Academy™ steering committee (co-authors SK and LDS) and reviewed by 13 experts/advisors and 2 sponsor personnel during 2 rounds of review. This was followed by an iterative Delphi-based review of proposed survey items to decide on the final survey instruments. The survey questions were chosen to cover the comprehensive range of diagnosis and treatment of CDH from the perspective of both program directors and fellowship graduates, and based on the model provided by the AAN. The surveys were delivered electronically to 2 separate populations of active sleep medicine fellowship program directors or associate program directors and recent sleep medicine fellowship graduates; program directors and graduates were not necessarily from the same programs. Each survey took approximately 15 min to complete. The surveys were administered between April 8, 2024, and July 9, 2024.

The surveys assessed perceptions among sleep medicine program directors and recent fellowship graduates on the quality of education for a specific subset of sleep disorders. The surveys were created by modeling trainee and program director surveys from the AAN [23–27]. Topics of interest included presenting features, diagnosis, testing, available treatments (including pharmacology, proper dosing, side effects, drug interactions, and polypharmacy), and management across the lifespan of CDH and in special patient populations (eg, pediatric patients).

The project was approved by Stanford University's institutional review board (IRB# 73428, approved April 2, 2024) and complied with all relevant laws and institutional guidelines. Participants provided informed consent to participate via email before they began answering survey questions.

2.3. Statistical analysis

Survey responses were summarized using standard descriptive statistics (eg, counts and percentage by level for categorical data; counts, mean, standard deviation, median, quartiles, minimum, and maximum for continuous data).

3. Results

Of the approximately 110 program directors who received surveys, 29 (≈26%) responded; of the approximately 740 post-fellowship graduates who were eligible to receive surveys through their program directors or other distribution channels, 48 (≈6%) responded (Table 1; Table 2). Post-fellowship graduates most commonly came from internal medicine backgrounds, whereas program directors primarily specialized in pulmonary/critical care medicine (Table 1; Table 2).

Overall, program directors reported that their graduating fellows are fully prepared to practice sleep medicine and manage CDH (Fig. 1). However, 2 (6.9%) program directors strongly disagreed that their graduating fellows are fully prepared to practice sleep medicine and manage CDH (Fig. 1).

Table 1
Demographic characteristics of program directors and post-fellowship graduates.

Characteristic	Program Directors (N = 29)	Post-Fellowship Graduates (N = 48)
Primary (nonsleep) training,[a] n (%)		
Ear, nose, and throat	0	2 (4.2)
Family medicine	2 (6.9)	6 (12.5)
Internal medicine	2 (6.9)	18 (37.5)
Neurology	9 (31.0)	6 (12.5)
Pediatrics	0	6 (12.5)
Pediatric subspecialty	0	4 (8.3)
Psychiatry	2 (6.9)	2 (4.2)
Pulmonary/critical care	15 (51.7)	11 (22.9)
Other		
Pediatric neurology	0	1 (2.1)
Pediatric Pulmonology	0	1 (2.1)
Sleep medicine	1 (3.4)	0
Gender (self-identified), n (%)		
Prefer not to answer	3 (10.3)	5 (10.4)
Man	16 (55.2)	19 (39.6)
Woman	10 (34.5)	23 (47.9)
Non-binary	0	1 (2.1)
Race (self-identified),[a] n (%)		
Prefer not to answer	4 (13.8)	6 (12.5)
American Indian or Alaska Native	0	1 (2.1)
Asian	7 (24.1)	17 (35.4)
Black or African American	0	1 (2.1)
White	18 (62.1)	24 (50.0)
Middle Eastern	0	3 (6.3)
Ethnicity (self-identified), n (%)		
Prefer not to answer	4 (13.8)	5 (10.4)
Not Hispanic or Latino	25 (86.2)	43 (89.6)

N/A, not available.
[a] Respondents could have chosen >1 answer; percentages may not add to 100%.

Table 2
Practice characteristics of program directors or post-fellowship graduates.

Characteristic	Program Directors (N = 29)
Role, n (%)	
Fellowship program director	24 (82.8)
Associate fellowship program director	5 (17.2)
Years in current role, mean (SD)	8.1 (6.2)
	Post-Fellowship Graduates (N=48)
Current career stage, n (%)	
≤1 year post sleep medicine fellowship	12 (25.0)
>1 to ≤2 years post sleep medicine fellowship	10 (20.8)
>2 to ≤3 years post sleep medicine fellowship	9 (18.8)
>3 to ≤4 years post sleep medicine fellowship	5 (10.4)
>4 to ≤5 years post sleep medicine fellowship	12 (25.0)
Current practice setting,[a] n (%)	
University based	16 (35.6)
Government based	4 (8.9)
Hospital based	15 (33.3)
Multispecialty group	5 (11.1)
Single-specialty group	4 (8.9)
Solo practice	2 (4.4)
Other public or private hospital/clinic	7 (15.6)
Other	
Children's hospital	1 (2.2)
No longer practicing	1 (2.2)

SD, standard deviation.
[a] Respondents could have chosen >1 answer; percentages may not add to 100%.

3.1. Training experience and adequacy in CDH

Program directors reported that, during fellowship training, the trainees saw a median of 15 (range: 2, 50) patients with narcolepsy type 1 (NT1), 20 (range: 3, 50) patients with narcolepsy type 2 (NT2), 20 (range: 2, 50) patients with idiopathic hypersomnia, and 1 (range: 0, 300) patient with recurrent hypersomnia. Program directors reported high levels of confidence in the adequacy of training in diagnosing and treating NT1 (96.6%), NT2 (100%), and idiopathic hypersomnia (100%), but relatively fewer had confidence about recurrent hypersomnia (82.8%) (Fig. 2).

During fellowship training, post-fellowship graduates reported having seen a median of 10 (range: 0, 100) patients with NT1, 10 (range: 1, 60) patients with NT2, 10 (range: 1, 50) patients with idiopathic hypersomnia, and no (range: 0, 20) patients with recurrent hypersomnia. Large proportions of post-fellowship graduates reported high levels of confidence in the adequacy of training in diagnosing and treating NT1 (90.9%), NT2 (90.9%), and idiopathic hypersomnia (86.4%), but relatively fewer had confidence about recurrent hypersomnia (59.1%) (Fig. 2).

3.2. Diagnostic tools

All program directors agreed or strongly agreed that fellowship training fully prepared trainees to use PSG (100%), MSLT (100%), and the ESS (100%) (Fig. 3). The majority of program directors (54.5%) reported that fellowship training fully prepared trainees to use actigraphy (Fig. 3). As reported by program directors, the median number of diagnostic tests interpreted by fellows during their training was 200 (IQR: 200, 250) for PSG, 50 (IQR: 30, 100) for home sleep testing, 25

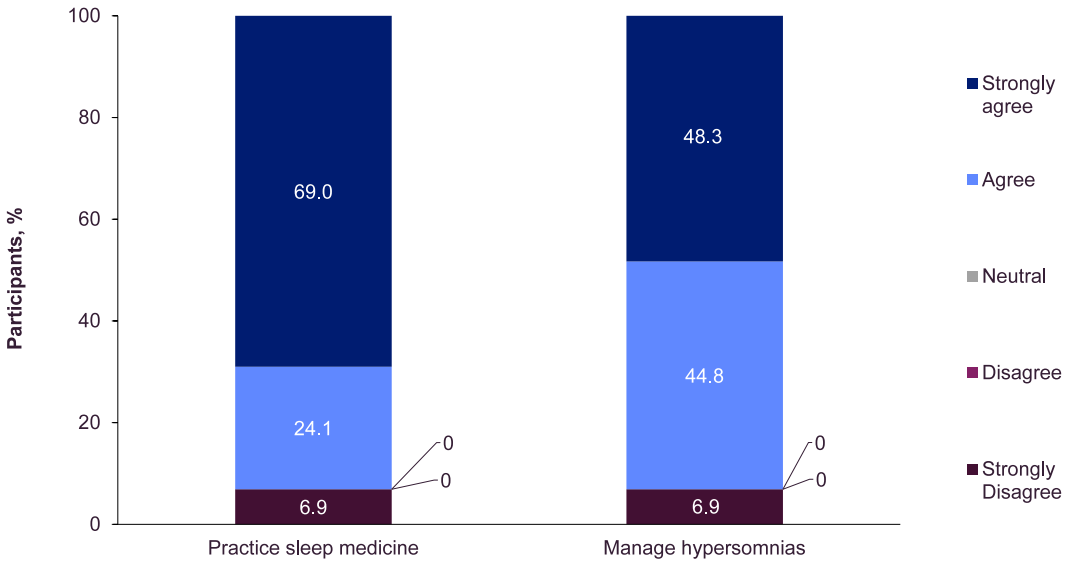


Fig. 1. Post-fellowship graduates' preparedness in sleep medicine and hypersomnia management reported by program directors.[a]
[a]Program directors' response to, "I feel my graduating fellows are fully prepared to practice sleep medicine/manage hypersomnias."

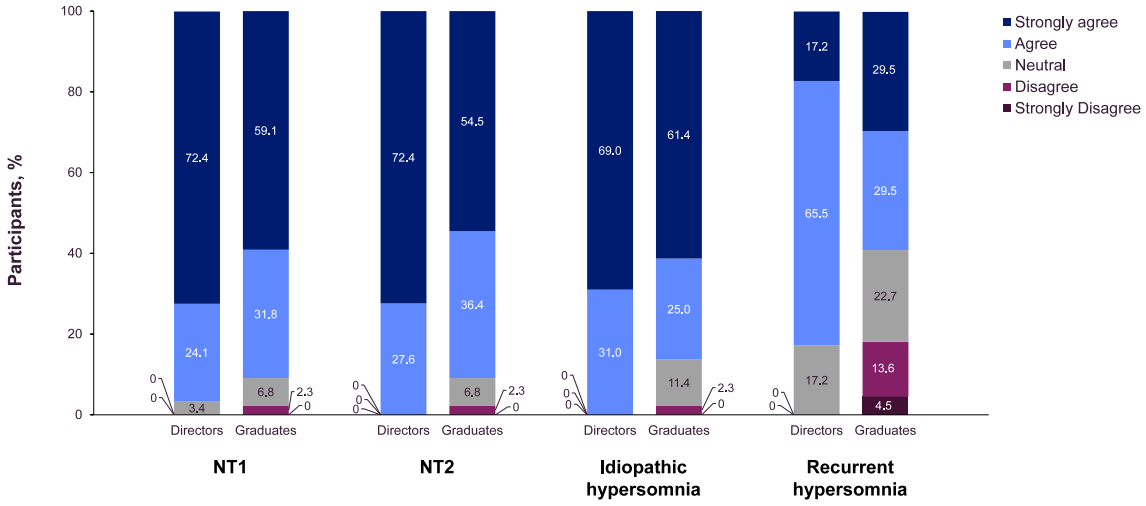


Fig. 2. Training adequacy in diagnosing and treating specific CDH reported by program directors[a] and post-fellowship graduates.[b]
[a]Program directors' response to, "Specific to central disorders of hypersomnolence, the program provides adequate training in diagnosing and treating NT1/NT2/idiopathic hypersomnia/recurrent hypersomnia."
[b]Post-fellowship graduates' response to, "Specific to central disorders of hypersomnolence, I feel comfortable diagnosing and treating NT1/NT2/idiopathic hypersomnia/recurrent hypersomnia."
CDH, central disorders of hypersomnolence; NT1, narcolepsy type 1; NT2, narcolepsy type 2.

(IQR: 14, 30) for MSLT, and 5 (IQR: 1, 10) for actigraphy.

All or nearly all post-fellowship graduates agreed or strongly agreed that fellowship training fully prepared them to use PSG (100%), MSLT (97.1%), and the ESS (100%) (Fig. 3). Fewer than half of the post-fellowship graduates (44.1%) reported that they felt fully prepared to use actigraphy (Fig. 3). As reported by post-fellowship graduates, the median number of diagnostic tests interpreted during fellowship training was 150 (IQR: 100, 200) for PSG, 50 (IQR: 10, 100) for home sleep testing, 20 (IQR: 15, 25) for MSLT, and 0 (IQR: 0, 8) for actigraphy.

3.3. Management and treatment of CDH

Program directors reported high levels of confidence regarding the adequacy of training in managing medication side effects (96.6%), stopping or transitioning medications (86.2%), managing multiple medications and polypharmacy (89.7%), and managing pediatric patients (72.4%) (Fig. 4). The number of medication prescriptions managed by fellows reported by program directors varied, with a median of 0 (IQR: 0, 0) for lithium, 10 (IQR: 4, 20) for methylphenidate, 5 (IQR: 0, 10) for oxybates, and 2 (IQR: 1, 5) for pitolisant. Program directors also reported uniformly high levels of confidence regarding the adequacy of training in treating patients with hypersomnia using therapies such as methylphenidate (100%) and oxybates (96.6%) (Fig. 5). Few program directors reported confidence with lithium (3.4%), whereas a much higher proportion had confidence with pitolisant (86.2%).

On average during their fellowship training, moderate proportions of post-fellowship graduates reported feeling confident about the adequacy of training in managing medication side effects (70.5%), stopping or transitioning medications (45.5%), managing multiple medications and polypharmacy (47.7%), and managing pediatric patients (34.1%) (Fig. 4). The number of medication prescriptions managed during

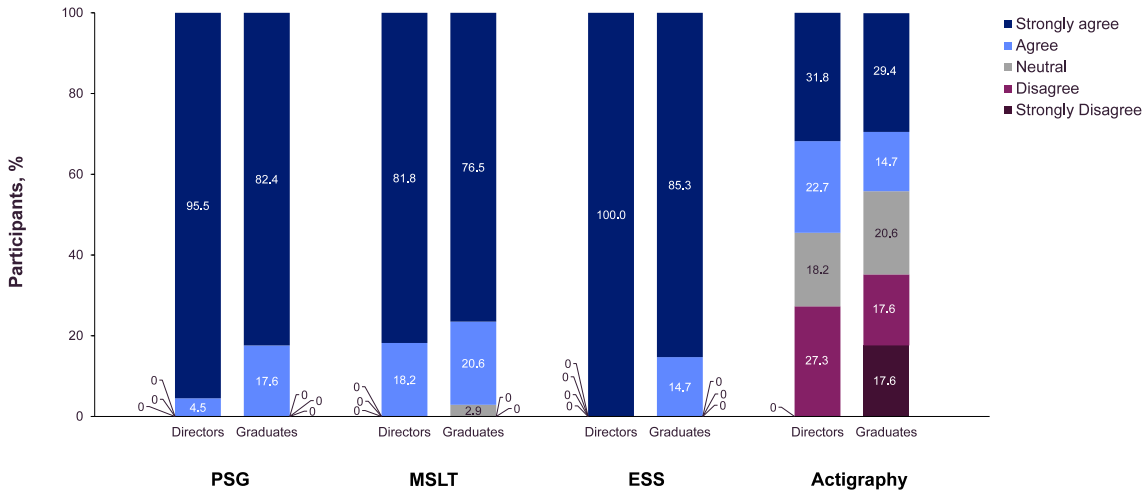


Fig. 3. Training adequacy in diagnostic tools reported by program directors[a] and post-fellowship graduates.[b]
[a]Program directors' response to, "On graduating, fellowship training fully prepares our trainees to use the following diagnostic tools."
[b]Post-fellowship graduates' response to, "Fellowship training fully prepared me to use the following diagnostic tools."
ESS, Epworth Sleepiness Scale; MSLT, Multiple Sleep Latency Test; PSG, polysomnography.

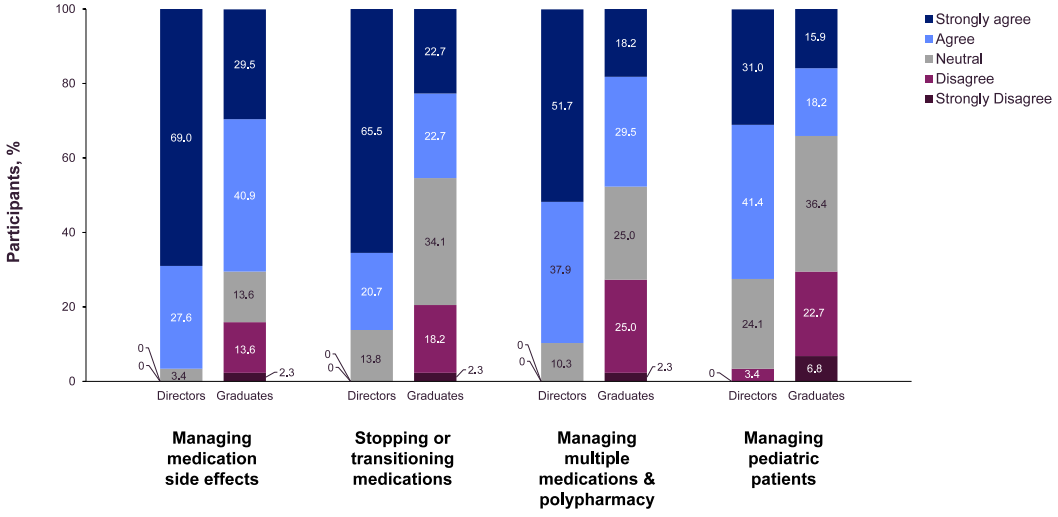


Fig. 4. Training adequacy in the medical management of CDH reported by program directors[a] and post-fellowship graduates.[b]
[a]Program directors' response to, "In the diagnosis and management of CDH, the program provides adequate training for managing medication side effects/stopping or transitioning medications/managing multiple medications and polypharmacy/managing pediatric patients."
[b]Post-fellowship graduates' response to, "In the diagnosis and management of CDH, my fellowship adequately prepared me for managing medication side effects/stopping or transitioning medications/managing multiple medications and polypharmacy/managing pediatric patients."
CDH, central disorders of hypersomnolence.

training reported by post-fellowship graduates varied, with a median of 0 (IQR: 0, 0) for lithium, 5 (IQR: 1, 10) for methylphenidate, 5 (IQR: 2, 20) for oxybates, and 1 (IQR: 0, 5) for pitolisant. Post-fellowship graduates reported the highest levels of confidence regarding the adequacy of training in treating patients with hypersomnia using medications such as methylphenidate (79.5%), oxybates (81.8%), and pitolisant (70.5%) (Fig. 5). Post-fellowship graduates reported a low level of confidence with lithium (6.8%; which was rarely prescribed [mean, 0.4 prescriptions per graduate]).

4. Discussion

The findings of this survey showed that both fellowship program directors and recent graduates reported high adequacy of overall sleep medicine training, particularly in the use of diagnostic tools such as PSG, MSLT, and ESS. This perception underscores the strengths of existing curricula across training programs. However, the findings also support the original hypothesis that potential educational gaps still remain regarding longitudinal care and developing competence in the evaluation and management of narcolepsy, idiopathic hypersomnia, and recurrent hypersomnia, as perceived by sleep medicine program directors and recent fellowship graduates.

There may be a potential correlation between perceived training adequacy in dealing with diagnoses/medications and clinical experience with CDH. That is, program directors likely had more experience diagnosing and treating patients with CDH than recent fellowship graduates, which could have contributed to some mismatch in perceptions of training adequacy between directors and post-fellowship graduates. Sleep medicine fellowships provide 1-year limited training in all sleep conditions, including rare disorders, which may be infrequently encountered during the training period. Consistent with this, survey results indicate low numbers of certain medication prescriptions (such

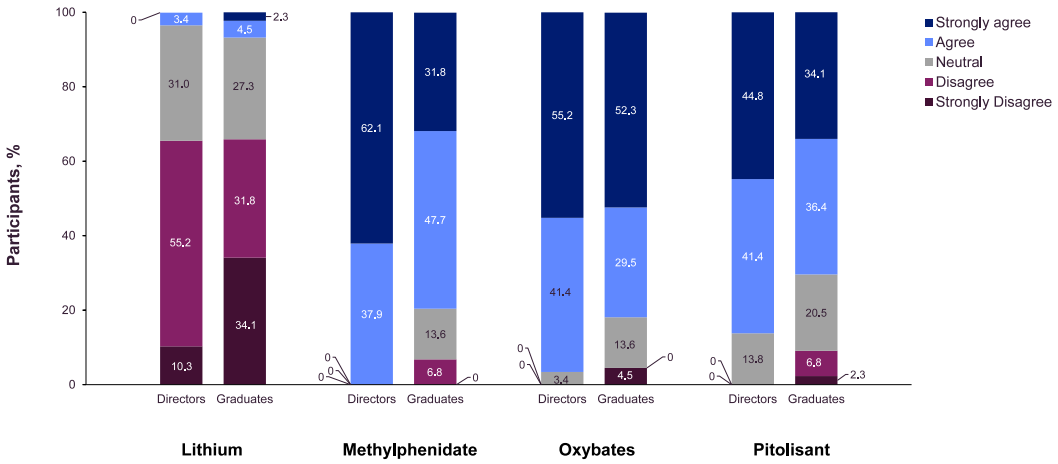


Fig. 5. Medication training adequacy reported by program directors[a] and post-fellowship graduates.[b]
[a]Program directors' response to, "The fellowship program provides adequate training in and exposure to management of central disorders of hypersomnolence with lithium/methylphenidate/oxybates/pitolisant."
[b]Post-fellowship graduates' response to, "I received adequate training during my fellowship program so I can confidently and appropriately treat patients with hypersomnia with lithium/methylphenidate/oxybates/pitolisant."

as lithium) managed by program directors and fellows for the treatment of patients with CDH. Furthermore, at some training programs, trainees may not be allowed to manage certain prescriptions, restricting their experience with prescribing medications. All of these findings can help institutions enhance their sleep medicine fellowship programs to better prepare fellowship graduates for independent practice. They may guide institutions and governing bodies in creating more specific, targeted guidelines and curricula for trainees and lifelong learners post fellowship. For example, because trainees are more likely to encounter common diagnoses earlier in their training, there may be a need for additional, extended training or enrichment toward the end of a fellowship to gain exposure to rarer cases.

Due to the nature of the survey, there are several limitations that restrict the ability to infer differences in the perceptions of sleep medicine fellowship training among program directors and recent fellowship graduates. First, program directors and post-fellowship graduates who responded did not necessarily represent the same sleep medicine program or cohort. Second, survey data were captured at the regional level to preserve privacy; however, sample sizes, which did not reach the target of 50 to 100 respondents in each cohort (particularly for program directors), were too small to explore practice pattern differences or similarities and also limited the generalizability of the survey results. Third, the low response rate among post-fellowship graduates limits the generalizability of these findings. Fourth, lack of complete institutional overlap between program directors and fellowship graduates constrains the ability to make comparisons between the 2 groups. Fifth, ascertainment or selection bias may have been present if, for example, satisfied program directors and/or dissatisfied fellowship graduates were overrepresented among respondents. Finally, variability in survey responses from program directors and fellowship graduates may have been caused by recall bias affecting the accuracy of reported data on using diagnostic tools, recognizing and managing sleep disorders, and treating patients with pharmacotherapy.

Notwithstanding those limitations, some discrepancies in the perceptions of training programs among program directors and post-fellowship graduates were noted. Compared with the responses of post-fellowship graduates, program directors reported marginally greater confidence in the adequacy of training in diagnosing and treating CDH and larger numbers of patients with CDH seen by fellows during fellowship training. Similarly, program directors reported marginally greater confidence in the adequacy of training in using diagnostic tools for CDH and larger numbers of sleep medicine diagnostics interpreted by fellows during fellowship training compared with the responses of post-fellowship graduates. Program directors also reported moderately greater confidence in the adequacy of training for the medical management of CDH (including managing medication side effects, stopping or transitioning medications, managing multiple medications and polypharmacy, and managing pediatric patients) compared with the responses of post-fellowship graduates. In addition, program directors reported marginally greater confidence in the adequacy of medication training (except for lithium) for the treatment of CDH compared with post-fellowship graduates.

Previously published studies on the perceptions of sleep medicine fellowship training have been primarily from the perspective of fellowship program directors [28,29]. However, surveys of neurology program directors and residents are available and have also shown a similar discordance between program directors and trainees regarding training adequacy [26,27]. As shown in the current analysis, such differences may underscore contrasting experiences and perspectives between the 2 groups and may also provide an opportunity to explore the relationship between training exposures and practice styles [30]. Nonetheless, the responses among recent fellowship graduates highlight at least some gaps in sleep medicine training experiences, based on those individuals reporting no ("0") exposure to managing patients with recurrent hypersomnia, interpreting actigraphy, and treating patients with lithium. Further surveys of fellowship graduates could explore details of training as they relate to complexities specific to CDH, such as differential diagnosis (including laboratory tests) among the numerous possible confounding causes of hypersomnolence (eg, poor sleep hygiene, other sleep disorders, and psychiatric disorders); practical application of pharmacotherapy (eg, determination of dosage, particular expected adverse effects, and appropriate use in clinical practice); and long-term prognosis and potential evolution of disease phenotype over time. Many of these topics have already been identified in professional reference documents [3,31,32]. Notwithstanding the potential value of such new information for guiding future educational objectives, the results from this study suggest that the main source of any shortcomings in current training fellowship curricula revolves around insufficient emphasis on longitudinal management.

There are several limitations to note besides those that were inherent to the design of the surveys. Analyses were entirely descriptive and lacked adjustment for the number of patients seen by program directors, who often have less clinical time than physicians who are not program directors. An important distinction was that the survey asked program directors about the adequacy of training, not the adequacy of their actual practice. There were few Black or African American respondents,

which may reflect minority underrepresentation in sleep fellowship programs. Due to small sample size and the descriptive nature of the analyses, this work did not explore whether there were effects of sex or gender on the findings.

5. Conclusions

The survey results highlight training adequacy in aspects of sleep medicine fellowship programs, particularly in diagnostic methods, but also suggest some gaps in CDH education as perceived by current program directors and recent fellowship graduates. These gaps indicate a need for new complements to standardized hypersomnia education. Programs such as Jazz Sleep Academy™, a collaborative educational initiative involving sleep medicine experts and fellowship program directors, may help address this need by providing multimodal educational resources and facilitating provider-to-provider education. Additional research is needed to further inform the creation of more specific, targeted guidelines and curricula to advance education in the diagnosis and management of CDH.

CRediT authorship contribution statement

Scott Kutscher: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Logan D. Schneider:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Sairam Parthasarathy:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Douglas S. Fuller:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation. **Nika Bejou:** Writing – review & editing. **Shawn Candler:** Writing – review & editing, Writing – original draft, Conceptualization.

Data sharing

All relevant data are provided with the manuscript and supporting files.

Author disclosures

S Kutscher has received personal compensation for serving on advisory boards and speakers bureaus for Jazz Pharmaceuticals.

LD Schneider has received personal compensation for serving on advisory boards and speakers bureaus for Jazz Pharmaceuticals, Eisai, and Avadel.

S Parthasarathy has received personal compensation for serving on an advisory board for Jazz Pharmaceuticals, Inc. He has served as a consultant for Apria Healthcare Inc., AbbVie, Inc., Fisher & Paykel, Inc., Zoll Medical Corp., and Apnimed, Inc.

DS Fuller and **N Bejou** are full-time employees of Jazz Pharmaceuticals who, in the course of this employment, have received stock options exercisable for, and other stock awards of, ordinary shares of Jazz Pharmaceuticals, plc.

S Candler is a former full-time employee of Jazz Pharmaceuticals who, in the course of this employment, received stock options exercisable for, and other stock awards of, ordinary shares of Jazz Pharmaceuticals, plc.

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Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that could be perceived to have influenced the work reported in this paper.

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